GURU NANAK DEV ENGINEERING COLLEGE

LUDHIANA | DEPARTMENT OF CIVIL ENGINEERING

ASSIGNMENT 1

CASE-BASED IN-CLASS QUIZ

Building Material and Testing | CCE101

B.Tech. Civil Engineering | Semester III | AY 2026–27

10 MARKS

20 MINUTES

Faculty: Dr. Yuvraj Singh | Department of Civil Engineering

GENERAL INSTRUCTIONS

- The quiz contains two comprehension passages, each followed by five MCQs.
- Section A assesses CO2; Section B assesses CO6. Each question carries 1 mark.
- There is no negative marking. Attempt all ten questions.
- Each passage and its questions will be displayed for a fixed duration.
- After moving to the next section, you cannot revisit the previous passage.
- Record one option (A, B, C or D) per question on your response sheet.
- Transfer your answers to the Google Form after completing both sections.
- The original response sheet will be collected after response transfer.
- Your Google Form responses must match your original recorded selections.
- Do not discuss answers. Correct answers will be discussed after the assessment.

SECTION A

CO2

Selection and Properties of Building Materials

Questions 1–5 | Maximum Marks: 5

Read carefully and record your responses on the response sheet.

COMPREHENSION PASSAGE 1

A civil engineer is selecting materials for a public building in a region with hot summers, seasonal rainfall and occasional damp conditions. The building includes reinforced concrete members, external masonry walls and a flat roof.

The engineer considers the following material properties:

- Concrete: Low permeability reduces water and harmful-substance ingress that can affect reinforcement durability.
- Masonry: Compressive strength matters for load-bearing use; water absorption matters in damp conditions.
- Aggregates: Sieve analysis determines particle-size distribution and grading for mix proportioning.
- Thermal insulation: Low thermal conductivity reduces heat transfer through walls and roofs.
- Fire resistance: Performance and its retention at elevated temperatures are important for fire safety.

Materials and tests must be selected for their intended applications rather than relying on one property for every situation.

QUESTION 1

Q1. Which property of concrete is most relevant for reducing the ingress of water and harmful substances?

A. High permeability

B. Low permeability

C. High water absorption

D. High surface smoothness

Select ONE option: A, B, C or D.

QUESTION 2

Q2. A masonry unit is proposed for a wall exposed to damp conditions. Which property should be assessed besides compressive strength?

A. Water absorption

B. Surface colour

C. Dimensional appearance alone

D. Paint compatibility

Select ONE option: A, B, C or D.

QUESTION 3

Q3. Which test determines the particle-size distribution of aggregates?

A. Slump test

B. Vicat test

C. Sieve analysis

D. Soundness test of cement

Select ONE option: A, B, C or D.

QUESTION 4

Q4. Which property is most desirable in a material intended to reduce heat transfer through an external wall?

A. High thermal conductivity

B. High water absorption

C. High density alone

D. Low thermal conductivity

Select ONE option: A, B, C or D.

QUESTION 5

Q5. Which consideration is most relevant when assessing the fire resistance of a building material?

A. Its colour under normal conditions

B. Its behaviour and retention of performance at elevated temperatures

C. Its surface smoothness before installation

D. Its water absorption alone

Select ONE option: A, B, C or D.

SECTION B

CO6

Application of IS Codes and Concrete Specifications

Questions 6–10 | Maximum Marks: 5

Read carefully and record your responses on the response sheet.

COMPREHENSION PASSAGE 2

A structural engineer is preparing concrete specifications for a reinforced concrete building. The engineer refers to IS 456:2000 for durability and structural concrete requirements and IS 10262:2019 for mix proportioning.

For this case study, the stated moderate-exposure durability requirements specify a maximum free water-cement ratio of 0.50 and a minimum concrete grade of M25.

The proposed mix has a free water-cement ratio of 0.52. Even if its trial strength satisfies the strength requirement, the mix must also meet the applicable durability requirements before acceptance.

Target mean compressive strength

$$\text{Target mean strength} = f_{ck} + 1.65\,S$$

For the separate M20 strength-calculation example, take fck = 20 MPa and S = 4 MPa. Target mean strength exceeds characteristic strength to account for expected production variations.

QUESTION 6

Q6. According to the passage, what is the maximum free water-cement ratio specified for the stated exposure condition?

A. 0.40

B. 0.45

C. 0.50

D. 0.55

Select ONE option: A, B, C or D.

QUESTION 7

Q7. The proposed mix has a free water-cement ratio of 0.52. What should the engineer do before approving the mix?

A. Accept it because the trial strength may be satisfactory

B. Revise the mix to comply with the specified maximum water-cement ratio

C. Ignore the exposure condition if the slump is satisfactory

D. Increase the water content to improve workability

Select ONE option: A, B, C or D.

QUESTION 8

Q8. For M20 concrete, calculate the target mean strength using the given expression and standard deviation.

A. 20.0 MPa

B. 24.0 MPa

C. 26.6 MPa

D. 30.0 MPa

Select ONE option: A, B, C or D.

QUESTION 9

Q9. Why is the target mean strength kept higher than the characteristic compressive strength?

A. To account for variations in concrete strength

B. To eliminate the need for curing

C. To ensure that every batch has identical strength

D. To avoid testing concrete specimens

Select ONE option: A, B, C or D.

QUESTION 10

Q10. An M20 mix meets the target mean strength but fails the applicable durability limit. Which decision is appropriate?

A. Approve the mix because strength is the only requirement

B. Accept the mix if its surface appears satisfactory

C. Disregard the durability limit for M20

D. Modify the mix to satisfy both strength and durability requirements

Select ONE option: A, B, C or D.

FINAL RESPONSE CHECK

Before submitting your response sheet:

- Ensure that Questions 1–10 each have one selected option.
- Check your recorded selections before submitting.
- Transfer the same responses to the Google Form.
- Submit your original response sheet when instructed.

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Scan the QR code to submit your responses



https://tinyurl.com/BMT-Assignment1-Quiz-2026

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